

Panel Data Models: OLS and GLS

ECMT 676 (Econometric II) Lecture Notes

TAMU

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- Suppose that, for each unit in the population, we have a (usually short) time series, $\{(\mathbf{x}_t, y_t) : t = 1, \dots, T\}$.
- A linear panel data model is

$$y_t = \mathbf{x}_t \boldsymbol{\beta} + u_t, \quad t = 1, \dots, T.$$

- In this setup, there is a single response variable, y_t , that we observe in several time periods.
- Having the same $\boldsymbol{\beta}$ for all t is not restrictive because $\mathbf{x}_t$ can be very general.
- It can (and usually should) include time-period dummies to allow a different intercept in each period. It can include interactions between time-period dummies and other variables to allow partial effects to change over time.

- We can write the model

$$\begin{aligned}y_1 &= \mathbf{x}_1\boldsymbol{\beta} + u_1 \\y_2 &= \mathbf{x}_2\boldsymbol{\beta} + u_2 \\&\vdots \\y_T &= \mathbf{x}_T\boldsymbol{\beta} + u_T\end{aligned}$$

- Suggests that, even though the applications are very different, a common statistical framework can be used for SUR (seeming unrelated regression) and panel data.
- $\mathbf{x}_t$ has the same dimension for all t .
- In SUR, the dimension of the covariates generally changes across equation.

EXAMPLE: Suppose that for the years 2000 through 2003 an annual family saving equation is

$$\begin{aligned} sav_t = & \beta_0 + \beta_1 d01_t + \beta_2 d02_t + \beta_3 d03_t \\ & + \beta_4 inc_t + \beta_5 size_t + \beta_6 educ_t + \beta_7 e401k_t \\ & + \beta_8 d01_t \cdot e401k_t + \beta_9 d02_t \cdot e401k_t + \beta_{10} d03_t \cdot e401k_t + u_t \end{aligned}$$

- A variable such as $educ_t$, education of the household head, might not change over time for many of the units.
- Some variables, such as gender, do not change over time for any units in the population.
- The general model allows variables that change only across time (such as year dummies), only across unit (such as gender), and across unit and time (typically, the most interesting variables).

EXAMPLE: A distributed lag model for county-level poverty rates might look like

$$\begin{aligned} poverty_t = & \alpha_t + \delta_0 welfare_t + \delta_1 welfare_{t-1} + \delta_2 welfare_{t-2} \\ & + \gamma_1 educ_t + \gamma_2 perc_children_t + \gamma_3 perc_elderly_t + u_t, \end{aligned}$$

where $welfare_t$ is per-capital welfare spending.

- In deciding how to estimate a model such as this, we must be very careful in stating exogeneity assumptions for the explanatory variables.
- Education and age distribution variables are just a couple of possible controls. The α_t are different period intercepts.
- The long run propensity is $\delta_0 + \delta_1 + \delta_2$, which is often easier to estimate than the individual δ_j .

- An attractive assumption in the general equation $y_t = \mathbf{x}_t\boldsymbol{\beta} + u_t$ is **contemporaneous exogeneity (CE)**:

$$E(u_t|\mathbf{x}_t) = 0, \quad t = 1, \dots, T, \quad (1)$$

which implies

$$E(y_t|\mathbf{x}_t) = \mathbf{x}_t\boldsymbol{\beta}, \quad t = 1, \dots, T.$$

- Rules out omitted variables, measurement error, and so on, but it does not restrict correlation between $\mathbf{x}_s$ and u_t for $s \neq t$.
- Stated as a zero conditional mean, it also implies correct functional form of $E(y_t|\mathbf{x}_t)$.

- An assumption that is stronger than CE, but can be justified when we assume the dynamics (with respect to the elements of $\mathbf{x}_t$) are completely specified, is **sequential exogeneity (SQE)**:

$$E(u_t | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_1) = 0, \quad t = 1, \dots, T, \quad (2)$$

- As we will see, it can be applied to distributed lag models and models with lagged dependent variables.
- Equation (2) implies

$$E(y_t | \mathbf{x}_t, \mathbf{x}_{t-1}, \dots, \mathbf{x}_1) = E(y_t | \mathbf{x}_t) = \mathbf{x}_t \boldsymbol{\beta},$$

which makes it clear that enough lags of whatever have been included in $\mathbf{x}_t$ so that no further lags are needed.

- An even stronger assumption, both technically and practically, is **strict exogeneity** (SE):

$$E(u_t | \mathbf{x}_1, \dots, \mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \dots, \mathbf{x}_T) = 0, t = 1, \dots, T.$$

This, of course, implies that $\mathbf{x}_s$ and u_t are uncorrelated for all s and t , including $s < t$, $s = t$, and $s > t$.

- In the context of $y_t = \mathbf{x}_t \boldsymbol{\beta} + u_t$, SE is the same as

$$E(y_t | \mathbf{x}_1, \dots, \mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_{t+1}, \dots, \mathbf{x}_T) = E(y_t | \mathbf{x}_t) = \mathbf{x}_t \boldsymbol{\beta},$$

where the first equality is the critical one. It says that, given whatever is in $\mathbf{x}_t$, the covariates from other time periods do not help explain y_t .

- The main concern with SE is that it rules out feedback from shocks to y at time t to future outcomes on $\mathbf{x}$.
- In particular, u_t and $\mathbf{x}_{t+1}$ cannot be correlated.
- An individual might become married or divorced in year $t + 1$ depending on the shock in time t .
- A school might adjust class size next year based on shocks to student performance this year.

EXAMPLE: A firm-level production function might look like

$$\log(Q_t) = \alpha_t + \beta_1 \log(K_t) + \beta_2 \log(L_t) + \beta_3 \log(M_t) + u_t,$$

where u_t contains productivity and other shocks.

- The SE assumption rules out the possibility that firms adjust inputs (capital, labor, materials) at, say, time $t + 1$ in reaction to shocks at time t .
- This example shows why maintaining fixed regressors in economic applications with panel data is a non-starter.
- We may or may not think strict or contemporaneous exogeneity hold for the production function, but at least the question makes sense when we view $\mathbf{x}_t$ as a random outcome along with u_t .

- In the poverty/welfare spending distributed lag example, we might be worried about “feedback effects.”
- For example, it might make sense to think of welfare payments being generated by something like

$$welfare_{t+1} = \eta_t + \lambda_1 poverty_t + r_{t+1},$$

which would violate SE.

- Any model with a lagged dependent variable *must* violate SE.
- For example, for a population of students,

$$score_t = \rho_1 score_{t-1} + \gamma_1 classsize_t + \gamma_2 faminc_t + u_t$$

- The reason for putting in the lagged test score is to control for the possibility that classroom assignment depends on prior performance.

- In general, write a model with a single lag as

$$\begin{aligned}y_t &= \rho_1 y_{t-1} + \mathbf{z}_t \gamma + u_t, \quad t = 1, 2, \dots, T \\ E(u_t | y_{t-1}, \mathbf{z}_t) &= 0\end{aligned}$$

Then $\mathbf{x}_t = (y_{t-1}, \mathbf{z}_t)$ and so

$$(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T) = (y_0, y_1, \dots, y_{T-1}, \mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_T)$$

Therefore, for $t < T$,

$$E(y_t | \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T) = y_t \neq E(y_t | \mathbf{x}_t)$$

- As with finite distributed lag models, for models with lagged y we often want SQE to hold – and that may or may not be true. But SE cannot hold.
- We discuss dynamic models in more detail later.

Pooled OLS Estimation

- We write the model for a random draw i as

$$\mathbf{y}_i = \mathbf{X}_i \boldsymbol{\beta} + \mathbf{u}_i$$

where $\mathbf{y}_i$ is $T \times 1$, $\mathbf{X}_i$ is $T \times K$, $\boldsymbol{\beta}$ is $K \times 1$, and $\mathbf{u}_i$ is $T \times 1$:

$$\mathbf{y}_i = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{pmatrix}, \quad \mathbf{X}_i = \begin{pmatrix} \mathbf{x}_{i1} \\ \mathbf{x}_{i2} \\ \vdots \\ \mathbf{x}_{iT} \end{pmatrix}, \quad \mathbf{u}_i = \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{pmatrix}$$

The observed data is $\{(\mathbf{X}_i, \mathbf{y}_i) : i = 1, \dots, N\}$, where N is the sample size.

- We want to estimate the population parameter vector $\boldsymbol{\beta}$.

POLS is consistent under contemporaneous exogeneity.

$$E(\mathbf{x}'_{it} u_{it}) = \mathbf{0}, t = 1, \dots, T.$$

Actually it is consistent under a weaker assumption.

$$\sum_{t=1}^T E(\mathbf{x}'_{it} u_{it}) = \mathbf{0}. \quad (3)$$

Rank condition:

$$\text{rank} \left(\sum_{t=1}^T E(\mathbf{x}'_{it} \mathbf{x}_{it}) \right) = \text{rank}(E\mathbf{X}'_i \mathbf{X}_i) = K.$$

- There are cases (later) where the rank condition does not hold for each t but it does hold summed across t .

Identification

$$\begin{aligned} E[\mathbf{X}'_i(\mathbf{y}_i - \mathbf{X}_i\beta)] &= \mathbf{0} \\ E(\mathbf{X}'_i \mathbf{X}_i)\beta &= E(\mathbf{X}'_i \mathbf{y}_i) \\ \beta &= [E(\mathbf{X}'_i \mathbf{X}_i)]^{-1} E(\mathbf{X}'_i \mathbf{y}_i) \end{aligned}$$

For exogeneity condition, we only need $E\mathbf{X}'_i \mathbf{u}_i = 0$ (which is (3), weaker than CE) here.

The POLS estimator looks just like the single-equations OLS estimator, but $\mathbf{X}_i$ is a matrix and $\mathbf{y}_i$ is a vector:

$$\begin{aligned}\hat{\beta}_{POLS} &= \left(N^{-1} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{x}'_i \mathbf{y}_i \right) \\ &= \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} \mathbf{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}'_{it} y_{it} \right),\end{aligned}$$

THEOREM: Under (3) and rank condition, POLS on a random sample is consistent:

$$\text{plim}_{N \rightarrow \infty} (\hat{\beta}_{POLS}) = \beta.$$

- Here we assume fixed T (number of time periods) with N growing. Reflects most micro panel data sets: many units (families, firms), few time periods per unit.
- The fixed T scenario is very convenient: it means the limit theorems are for random sampling (as in an introductory probability and statistics course).
- Further, the results allow *any* kind of time series dependence in the explanatory variables or errors.
- The case $T \rightarrow \infty$ is more in the realm of time series econometrics and would require restricting the time series correlation to get unified results.

Asymptotic Normality and Inference

- The proof of asymptotic normality is similar to the single-equation case, but we have to be more careful with the linear algebra. Write

$$\sqrt{N}(\hat{\beta} - \beta) = \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \mathbf{u}_i \right).$$

- By the CLT for i.i.d. random vectors,

$$N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \mathbf{u}_i \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{B})$$

$$\mathbf{B} = \text{Var}(\mathbf{x}_i' \mathbf{u}_i) = E(\mathbf{x}_i' \mathbf{u}_i \mathbf{u}_i' \mathbf{x}_i).$$

- Under homoskedasticity, the asymptotic variance does not have a simpler form (still assuming a sandwich form).
- This is in contrast to the single-equation model.

- Define

$$\mathbf{A} = E(\mathbf{X}_i' \mathbf{X}_i).$$

$$\begin{aligned} N^{1/2}(\hat{\beta} - \beta) &= \mathbf{A}^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \right) \\ &\quad + \left[\left(N^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \right)^{-1} - \mathbf{A}^{-1} \right] \left(N^{-1/2} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \right) \\ &= \mathbf{A}^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \right) + o_p(1) \cdot O_p(1). \end{aligned}$$

- So

$$N^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}).$$

$$\hat{\mathbf{A}} = N^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{X}_i \quad (\text{a } K \times K \text{ matrix})$$

- The SOLS residuals are

$$\mathbf{u}_i = \mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}} \quad (\text{each is } G \times 1)$$

- A fully robust estimator of $\mathbf{B}$ – that is, an estimator valid under SOLS.1 and SOLS.2, without any second moment assumptions on $\mathbf{u}_i$ – is

$$\hat{\mathbf{B}} = (N - K)^{-1} \sum_{i=1}^N \mathbf{X}_i' \mathbf{u}_i \mathbf{u}_i' \mathbf{X}_i.$$

- The degrees-of-freedom adjustment is common but unnecessary with large N . The resulting sandwich estimator is

$$\widehat{Avar}(\hat{\beta}) = \frac{N}{(N-K)} \left(\sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}'_i \mathbf{u}_i \mathbf{u}'_i \mathbf{x}_i \right) \left(\sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1}. \quad (4)$$

- It allows the variances, $Var(u_t)$, and covariances, $Cov(u_t, u_s)$, to be unrestricted. Thus, it is robust to arbitrary heteroskedasticity as a function of time and to any serial correlation pattern in the errors, $\{u_{it} : t = 1, \dots, T\}$. In addition, it is robust to the conditional variances and covariances being any function of $\{\mathbf{x}_{it} : t = 1, \dots, T\}$.

Testing

- Obtain asymptotic standard errors from the sandwich variance estimator to construct large-sample t statistics (which converge to standard normal under the null as $N \rightarrow \infty$) and confidence intervals.
- Consider hypotheses

$$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r},$$

where $\mathbf{R}$ is $Q \times K$, $\mathbf{r}$ is $Q \times 1$, $Q \leq K$. Most convenient is the **Wald statistic**:

$$\mathcal{W} = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})'(\mathbf{R}\hat{\mathbf{V}}\mathbf{R}')^{-1}(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \overset{d}{\sim} \chi_Q^2$$

under H_0 , where $\hat{\mathbf{V}}$ is given in (4).

The airfare example: route-level data (San Diego-Boston, Seattle-Miami,...)

$$lfare = \beta_0 + \beta_1 lconcen + \beta_2 ldist + \beta_3 ldistsq + \beta_4 y98 + \beta_5 y99 + \beta_6 y00 + u$$

It is important to control for distance (short route has large concentration ratio)
lfare: average one-way fare over the year (so no date-effect needs to worry about)

- By “generalized least squares” we mean exploiting different *unconditional* variances across equation (time, in the panel data case) and nonzero *unconditional* covariances across equations.
- Here we do not explore cases where the conditional variance-covariance matrix is a function of $\mathbf{X}_i$.
- Write the equation in system form (for a random draw i) as

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{u}_i$$

where $\mathbf{y}_i$ is $T \times 1$, $\mathbf{X}_i$ is $T \times K$, and $\mathbf{u}_i$ is $T \times 1$.

- The $T \times T$ *unconditional* variance-covariance matrix plays a key role.

$$\Omega \equiv E(\mathbf{u}_i \mathbf{u}_i') = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1G} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2G} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1G} & \sigma_{2G} & \cdots & \sigma_G^2 \end{pmatrix}.$$

- Assume $E(\mathbf{u}_i) = \mathbf{0}$, which is true in almost all applications; it is no assumption at all when the various equations contain an intercept.
- If $\mathbf{u}_i$ does not have a zero mean then Ω is not the V-C matrix, but everything goes through.

- We already discussed system OLS. What else might we do?
- Without additional assumptions, suppose we use a “generalized least squares” transformation. Assume that we know Ω .
- Transform the equation to remove correlations in errors and make variances constant (actually, unity):

$$\underbrace{\Omega^{-1/2}\mathbf{y}_i}_{\mathbf{y}_i^*} = \underbrace{\Omega^{-1/2}\mathbf{X}_i}_{\mathbf{X}_i^*}\boldsymbol{\beta} + \underbrace{\Omega^{-1/2}\mathbf{u}_i}_{\mathbf{u}_i^*}.$$

- Note how Ω is assumed to be nonsingular. $\Omega^{-1/2}$ is a symmetric, nonsingular matrix such that $\Omega^{-1/2}\Omega^{-1/2} = \Omega^{-1}$ and $\Omega^{-1/2}\Omega\Omega^{-1/2} = \mathbf{I}_G$.
- Then

$$E(\mathbf{u}_i^* \mathbf{u}_i^{*'}) = \Omega^{-1/2} E(\mathbf{u}_i \mathbf{u}_i') \Omega^{-1/2} = \mathbf{I}_G.$$

- Now apply System OLS to the transformed equation

$$\mathbf{y}_i^* = \mathbf{X}_i^* \boldsymbol{\beta} + \mathbf{u}_i^*$$

to get the GLS estimator of $\boldsymbol{\beta}$.

- The GLS estimator is

$$\begin{aligned}\beta^* &= \left(\sum_{i=1}^N \mathbf{x}_i^{*'} \mathbf{x}_i^* \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i^{*'} \mathbf{y}_i^* \right) \\ &= \left(\sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{y}_i \right).\end{aligned}$$

- To study the consistency of GLS, substitute $\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{u}_i$ and simplify:

$$\begin{aligned}\boldsymbol{\beta}^* &= \left(\sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} (\mathbf{x}_i \boldsymbol{\beta} + \mathbf{u}_i) \right) \\ &= \boldsymbol{\beta} + \left(\sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} \mathbf{u}_i \right) \\ &= \boldsymbol{\beta} + \left(N^{-1} \sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{x}'_i \Omega^{-1} \mathbf{u}_i \right).\end{aligned}$$

- By the law of large numbers,

$$N^{-1} \sum_{i=1}^N \mathbf{X}_i' \Omega^{-1} \mathbf{X}_i \xrightarrow{P} E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i),$$

where the matrices are $K \times K$.

- We need the rank condition that $E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i)$ is nonsingular.
- Then, by the LLN applied to $N^{-1} \sum_{i=1}^N \mathbf{X}_i' \Omega^{-1} \mathbf{u}_i$, consistency of β^* holds if

$$E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) = \mathbf{0}. \quad ((1.5))$$

- Important: In general, $E(\mathbf{X}'_i \Omega^{-1} \mathbf{u}_i) = \mathbf{0}$ is *not* implied by SOLS.1, that is,

$$E(\mathbf{X}'_i \mathbf{u}_i) = \mathbf{0}. \quad ((1.6))$$

- GLS transforms the orthogonality conditions; consequently, it may not be consistent when SOLS is.
- The orthogonality condition $E(\mathbf{X}'_i \Omega^{-1} \mathbf{u}_i) = \mathbf{0}$ is specific to the structure of Ω . A stronger condition is useful for several reasons.

Assumption SGLS.1 (Exogeneity):

$$E(\mathbf{X}_i \otimes \mathbf{u}_i) = \mathbf{0}. \quad \square \quad ((1.7))$$

- The use of the Kronecker product means that every element of $\mathbf{X}_i$ is uncorrelated with every element of $\mathbf{u}_i$, so any linear combination of $\mathbf{X}_i$ is uncorrelated with $\mathbf{u}_i$. In particular, $E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) = \mathbf{0}$ holds.
- In special cases $E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) = \mathbf{0}$ can hold when $E(\mathbf{X}_i \otimes \mathbf{u}_i) \neq \mathbf{0}$, but SGLS.1 implies that a variety of GLS estimators, even with a misspecified variance matrix, will be consistent.
- Shortly we will rely on SGLS.1 to justify ignoring estimation of Ω ; $E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) = \mathbf{0}$ is not sufficient.

- We already saw the appropriate rank condition:

Assumption SGLS.2 (Rank Condition): Ω is nonsingular and $E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i)$ is nonsingular. $\square$

THEOREM: Under SGLS.1 and SGLS.2, β^* is consistent for β as $N \rightarrow \infty$. $\square$

- We Must take the distinction between SGLS.1 and SOLS.1 seriously. If only $E(\mathbf{X}_i' \mathbf{u}_i) = \mathbf{0}$ holds, GLS is generally inconsistent.

- Asymptotic normality of GLS is also straightforward:

$$\begin{aligned}
 \sqrt{N}(\beta^* - \beta) &= \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{u}_i \right) \\
 &= \mathbf{A}^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{u}_i \right) \\
 &\quad + \left[\left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{x}_i \right)^{-1} - \mathbf{A}^{-1} \right] \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{u}_i \right) \\
 &= \mathbf{A}^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{u}_i \right) + o_p(1) \quad ((1.10))
 \end{aligned}$$

- Now define

$$\mathbf{A} = E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i). \quad ((1.11))$$

- Using the same argument as for SOLS,

$$\sqrt{N}(\boldsymbol{\beta}^* - \boldsymbol{\beta}) \xrightarrow{d} \text{Normal}(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}) \quad ((1.12))$$

$$\mathbf{B} = \text{Var}(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) = E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i \mathbf{u}_i' \Omega^{-1} \mathbf{X}_i) \quad ((1.13))$$

- Important: At this point, we cannot simplify $\mathbf{B}$ further. We are not assuming $\mathbf{X}_i$ is nonrandom, and we have not imposed enough assumptions on the joint distribution of $(\mathbf{X}_i, \mathbf{u}_i)$ [or the conditional distribution $D(\mathbf{u}_i | \mathbf{X}_i)$] to reduce $\mathbf{B}$.
- One consequence of the complicated expression for $\mathbf{B}$: under SGLS.1, SOLS.2, and SGLS.2, GLS need *not* be more efficient than SOLS. (Both are consistent and $\sqrt{N}$ -asymptotically normal.)

Feasible GLS

- Now we study the estimator from the previous section but where an estimator of Ω is used in place of Ω .
- Generally, let $\hat{\Omega}$ be a $T \times T$ matrix such that

$$\text{plim}_{N \rightarrow \infty} \hat{\Omega} = \Omega. \quad ((2.1))$$

- This assumption only makes sense when Ω has fixed dimension. (In the panel data case, T is fixed.)
- We almost always use

$$\hat{\Omega} = N^{-1} \sum_{i=1}^N \check{u}_i \check{u}_i' \quad ((2.2))$$

where $\check{u}_i \equiv \mathbf{y}_i - \mathbf{X}_i \check{\beta}$ are the $T \times 1$ POLS residuals ($\check{\beta}$ is the POLS estimator).

- Write

$$\begin{aligned}\check{u}_i &= \mathbf{y}_i - \mathbf{X}_i\boldsymbol{\beta} - \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{u}_i - \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta}) \\ \check{u}_i\check{u}_i' &= \mathbf{u}_i\mathbf{u}_i' - \mathbf{u}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathbf{X}_i' - \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})\mathbf{u}_i' \\ &\quad + \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathbf{X}_i'.\end{aligned}\tag{((2.3))}$$

$$\begin{aligned}N^{-1}\sum_{i=1}^N \check{u}_i\check{u}_i' &= N^{-1}\sum_{i=1}^N \mathbf{u}_i\mathbf{u}_i' - \underbrace{N^{-1}\sum_{i=1}^N \mathbf{u}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathbf{X}_i'}_{=O_p(N^{-1})} - \underbrace{N^{-1}\sum_{i=1}^N \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})\mathbf{u}_i'}_{=O_p(N^{-1})} \\ &\quad + \underbrace{N^{-1}\sum_{i=1}^N \mathbf{X}_i(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})(\check{\boldsymbol{\beta}} - \boldsymbol{\beta})'\mathbf{X}_i'}_{=O_p(N^{-1})}.\end{aligned}$$

- From this expression we can show that

$$\hat{\Omega} = N^{-1} \sum_{i=1}^N \mathbf{u}_i \mathbf{u}_i' + o_p(1). \quad ((2.4))$$

- We can show something stronger under SGLS.1 and SOLS.2:

$$\sqrt{N} \left(\hat{\Omega} - N^{-1} \sum_{i=1}^N \mathbf{u}_i \mathbf{u}_i' \right) = o_p(1). \quad ((2.5))$$

- The asymptotic equivalence in (2.5) is handy for performing inference about the elements of Ω : we can ignore the estimation error in $\check{\beta}$ (in large samples). Very useful for testing zero covariances, constant variances, and no serial correlation.
- Equation (2.5) does not go through under SOLS.1, although (2.4) does.

- The feasible GLS (FGLS) estimator is

$$\begin{aligned}\hat{\beta} &= \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{y}_i \right) \\ &= \beta + \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{u}_i \right)\end{aligned}\quad ((2.6))$$

Write

$$N^{-1} \sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i - N^{-1} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{x}_i = N^{-1} \sum_{i=1}^N \mathbf{x}_i' (\hat{\Omega}^{-1} - \Omega^{-1}) \mathbf{x}_i.$$

- Fact from matrix algebra: For conformable matrices **A**, **B**, and **C**,

$$\text{vec}(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A})\text{vec}(\mathbf{B})$$

where “vec” is the vectorization of a matrix (stacking the columns to make a long column vector).

- Under SGLS.1, SGLS.2, and $\hat{\Omega} = N^{-1} \sum_{i=1}^N \mathbf{u}_i \mathbf{u}_i' + o_p(1)$,

$$\begin{aligned} \text{vec} \left[N^{-1} \sum_{i=1}^N \mathbf{x}_i' (\hat{\Omega}^{-1} - \Omega^{-1}) \mathbf{x}_i \right] &= \left[N^{-1} \sum_{i=1}^N (\mathbf{x}_i' \otimes \mathbf{x}_i') \right] \text{vec}(\hat{\Omega}^{-1} - \Omega^{-1}) \\ &= O_p(1) \cdot o_p(1) \\ &= o_p(1). \end{aligned}$$

- Further,

$$\begin{aligned} N^{-1/2} \sum_{i=1}^N (\mathbf{X}_i' \hat{\Omega}^{-1} \mathbf{u}_i - \mathbf{X}_i' \Omega^{-1} \mathbf{u}_i) &= N^{-1/2} \sum_{i=1}^N \mathbf{X}_i' (\hat{\Omega}^{-1} - \Omega^{-1}) \mathbf{u}_i \\ &= \left[N^{-1/2} \sum_{i=1}^N (\mathbf{u}_i \otimes \mathbf{X}_i)' \right] \text{vec}(\hat{\Omega}^{-1} - \Omega^{-1}) \\ &= O_p(1) \cdot o_p(1) = o_p(1). \end{aligned}$$

- Combining the above two results gives

$$\begin{aligned}\sqrt{N}(\hat{\beta} - \beta) &= \left(N^{-1} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{x}_i \right)^{-1} \left(N^{-1/2} \sum_{i=1}^N \mathbf{x}_i' \Omega^{-1} \mathbf{u}_i \right) + o_p(1) \\ &= \sqrt{N}(\beta^* - \beta) + o_p(1).\end{aligned}\quad ((2.8))$$

- By the asymptotic equivalence lemma, the asymptotic distribution of $\sqrt{N}(\hat{\beta} - \beta)$ is the same as that of $\sqrt{N}(\beta^* - \beta)$.
- When

$$\sqrt{N}(\hat{\beta} - \beta^*) = o_p(1) \quad ((2.9))$$

we say that $\hat{\beta}$ and β^* are “asymptotically equivalent” or, more precisely, “ $\sqrt{N}$ -equivalent,” which is much stronger than saying that they are both consistent. (Under SGLS.1, SOLS.2, and SGLS.2, FGLS and SOLS are both consistent but they are not asymptotically equivalent.)

- It is not always true that first-stage estimation of population parameters can be ignored in a second stage (for example, see the control function slides). But, in this case, for $\sqrt{N}$ -asymptotics, we can treat FGLS as if it is GLS.
- If N is “small,” the statistical properties of $\hat{\beta}$ and β^* could be very different (and we would not know, in general, because β^* is infeasible).
- FGLS is not generally unbiased under $E(\mathbf{u}_i|\mathbf{X}_i) = \mathbf{0}$; GLS is (conditionally) unbiased (if the moments exist).

- It is important to remember that the following conditions are listed from weakest to strongest:

$$E(\mathbf{X}_i' \mathbf{u}_i) = \mathbf{0} \text{ (SOLS.1)}$$

$$E(\mathbf{X}_i \otimes \mathbf{u}_i) = \mathbf{0} \text{ (SGLS.1)}$$

$$E(\mathbf{u}_i | \mathbf{X}_i) = \mathbf{0} \text{ (Zero Conditional Mean)}$$

- By far the strongest assumption (which we have no use for) is $\mathbf{u}_i$ independent of $\mathbf{X}_i$ with $E(\mathbf{u}_i) = \mathbf{0}$.

- A fully robust sandwich variance matrix estimator can be used under SGLS.1 and SGLS.2. Let $\mathbf{u}_i \equiv \mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}$ be the FGLS residuals. Then

$$\widehat{\text{Avar}}(\hat{\boldsymbol{\beta}}) = \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{u}_i \mathbf{u}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1} ((2.10)) \\ \cdot \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1},$$

and sometimes with a degrees-of-freedom adjustment, such as multiplication by $N/(N - K)$.

- This estimator is robust to “system heteroskedasticity.” In other words, expression (2.10) allows the variance-covariance matrix of $\mathbf{u}_i$ conditional on $\mathbf{X}_i$ to depend arbitrarily on $\mathbf{X}_i$.
- Equation (2.10) extends the usual heteroskedasticity-robust variance matrix estimator to the system setting.

When is the “Usual” Variance Matrix Estimator for FGLS Valid?

Assumption SGLS.3 (System Homoskedasticity):

$$E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i \mathbf{u}_i' \Omega^{-1} \mathbf{X}_i) = E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i). \quad \square \quad ((2.11))$$

- In addition to essentially requiring $E(\mathbf{u}_i \mathbf{u}_i') = \Omega$, this condition implies that all squares and cross products, u_{ig}^2 , $u_{ig} u_{ih}$, are uncorrelated with the levels, squares, and cross products of elements in $\mathbf{X}_i$.

- SGLS.3 simply says that $\mathbf{B} = \mathbf{A}$, which means we can use

$$\widehat{\text{Avar}}(\hat{\beta}) = \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1}, \quad ((2.12))$$

which is the nonrobust (“usual”) FGLS variance matrix estimator.

- Sufficient for SGLS.3:

$$E(\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i) = E(\mathbf{u}_i \mathbf{u}_i'). \quad ((2.13))$$

- Use the law of iterated expectations:

$$\begin{aligned} E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i \mathbf{u}_i' \Omega^{-1} \mathbf{X}_i) &= E[E(\mathbf{X}_i' \Omega^{-1} \mathbf{u}_i \mathbf{u}_i' \Omega^{-1} \mathbf{X}_i | \mathbf{X}_i)] \\ &= E[\mathbf{X}_i' \Omega^{-1} E(\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i) \Omega^{-1} \mathbf{X}_i] \\ &= E(\mathbf{X}_i' \Omega^{-1} \Omega \Omega^{-1} \mathbf{X}_i) = E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i). \end{aligned}$$

When is FGLS more efficient than SOLS?

- Suppose we start with SGLS.1, $E(\mathbf{X}_i \otimes \mathbf{u}_i) = \mathbf{0}$ (which implies SOLS.1), add the two rank conditions, SOLS.2 and SGLS.2, and a particular form of the system homoskedasticity assumption:

$$E(\mathbf{u}_i \mathbf{u}_i' | \mathbf{X}_i) = E(\mathbf{u}_i \mathbf{u}_i'). \quad ((2.18))$$

It can be shown that

$$\text{Avar}(\hat{\beta}_{FGLS}) = [E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i)]^{-1} / N \quad ((2.19))$$

$$\begin{aligned} \text{Avar}(\hat{\beta}_{SOLS}) &= [E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} E(\mathbf{X}_i' \mathbf{u}_i \mathbf{u}_i' \mathbf{X}_i) [E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} / N \\ &= [E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} E(\mathbf{X}_i' \Omega \mathbf{X}_i) [E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} / N \quad ((2.20)) \end{aligned}$$

- Claim:

$$[E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} E(\mathbf{X}_i' \Omega \mathbf{X}_i) [E(\mathbf{X}_i' \mathbf{X}_i)]^{-1} - [E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i)]^{-1} \quad ((2.21))$$

is positive semi-definite.

- Write the difference as $\mathbf{C} - \mathbf{D}$. Then it suffices to show $\mathbf{D}^{-1} - \mathbf{C}^{-1}$ is p.s.d., that is

$$E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i) - E(\mathbf{X}_i' \mathbf{X}_i) [E(\mathbf{X}_i' \Omega \mathbf{X}_i)]^{-1} E(\mathbf{X}_i' \mathbf{X}_i) \quad ((2.22))$$

is p.s.d.

- Use the following trick. Let $\mathbf{Z}_i \equiv \Omega^{-1/2} \mathbf{X}_i$ and $\mathbf{W}_i \equiv \Omega^{1/2} \mathbf{X}_i$. Then $E(\mathbf{Z}_i' \mathbf{Z}_i) = E(\mathbf{X}_i' \Omega^{-1} \mathbf{X}_i)$, $E(\mathbf{W}_i' \mathbf{W}_i) = E(\mathbf{X}_i' \Omega \mathbf{X}_i)$, and $E(\mathbf{Z}_i' \mathbf{W}_i) = E(\mathbf{X}_i' \mathbf{X}_i)$. Therefore, the difference in (2.22) is

$$E(\mathbf{Z}_i' \mathbf{Z}_i) - E(\mathbf{Z}_i' \mathbf{W}_i) [E(\mathbf{W}_i' \mathbf{W}_i)]^{-1} E(\mathbf{Z}_i' \mathbf{W}_i), \quad ((2.23))$$

which looks like a matrix sum of squared residuals in the population. In fact, if we define the matrix residuals $\mathbf{R}_i = \mathbf{Z}_i - \mathbf{W}_i \Pi$ with $\Pi \equiv [E(\mathbf{W}_i' \mathbf{W}_i)]^{-1} E(\mathbf{Z}_i' \mathbf{W}_i)$, then it is easily seen that (2.23) is $E(\mathbf{R}_i' \mathbf{R}_i)$, which is necessarily p.s.d.